

Quantifying Roadway Hazards

Nevada County uses Ladrís AI to rank mitigation priorities

A NEED TO QUANTIFY HAZARDS

Nevada County is a rural county in the foothills of California's Sierra Nevada which contains an extensive Wildland Urban Interface (WUI). In late 2022, County OES needed a way to clearly quantify evacuation mitigation priorities for every road, neighborhood, and local community in the entire County, in compliance with AB747 and CA99.

SOLVED BY SCALING HUMAN DOMAIN EXPERTISE WITH AI

First, neighborhoods and potential shelter locations were determined by County OES. This information was used to generate scenarios with Ladrís AI, simulating conditions for 156,000 road segments to determine which exact roadways posed the greatest risk under hundreds of different potential evacuation scenarios. The top five neighborhoods were then ranked by worst case egress times, giving the County a clear roadmap to prioritize evacuation issue mitigation efforts.

OUTCOMES

Ranked Roadway Hazard Priorities

Nevada County quantified the current state of egress risk for the top 13 neighborhoods identified by its emergency managers, giving future mitigation efforts an obvious, measurable baseline from which to improve.

Local Knowledge Rapidly Scaled County-Wide

Nevada County's EMs were able to scale their local knowledge of evacuation threats rapidly; Using Ladrís AI, They answered within days a question that would have taken months using their previous approach to the work.

Set Roadmap for Ongoing Mitigation Efforts

Nevada County's transportation commission has continued to build off this work, evaluating the potential impacts of additional state highway capacity & making the case for mitigation spend to deliver improvements.

AT A GLANCE

Challenges

- Needed to prioritize County-wide roadway risks
- Limited time and resources
- Required quantitative data

Benefits

- Produced definitive rankings for roadway hazard mitigation
- Provided a force multiplier to County emergency managers
- Established a roadmap for future mitigation efforts & grants



"The data modeling aids our team beyond just evacuation planning; it also provides important data that can be used for fuel removal planning along evacuation routes and help identify potential infrastructure improvements."

Steve Monaghan

Chief Information Officer
County of Nevada